

Zhaode Wang

AI Inference Engine Expert · On-Device LLM · High-Performance Computing

Email: hi@zhaode.wang · github.com/wangzhaode

Education

Institute of Computing Technology, Chinese Academy of Sciences	M.S. in Computer Architecture	2017 - 2020
State Key Laboratory of Computer Architecture		
Shandong University	B.S. in Computer Science and Technology	2013 - 2017

Experience

Alibaba · Taotian Group	Senior Technical Expert · MNN Tech Lead	2020 - Present
MNN core architecture: Lead MNN architecture and heterogeneous performance optimization across ARM CPUs, Metal/OpenCL/Vulkan GPUs, and Hexagon/CoreML/NNAPI accelerators, covering operator kernels, memory planning, graph optimization and fusion, and model conversion; grew MNN to 15.8K+ GitHub stars. On-device AI platform: Built an embedded Python runtime and end-to-end deployment stack, enabling dozens of algorithms for Pailitao visual search, Taobao Live, security, and feed; supported Quark, Qwen, DingTalk, Youku, Taobao Instant Commerce, and Amap. MNN-LLM & production: Initiated and currently lead MNN-LLM for 100+ language and multimodal models, spanning 2/3/4/8-bit and mixed-precision quantization, KV-cache and attention optimization, and speculative decoding; deployed on-device LLMs in Taobao and Xianyu search and recommendation at tens-of-millions-user scale. On-device SLM: Led a team to train a 0.3B SLM from scratch for on-device applications, covering data collection and quality assessment, pre-training, SFT, RL, and end-to-end model evaluation.		
Cambricon Technologies	Compiler Engineer Intern	2018 - 2020
Developed compiler and linker components for Cambricon's BANG toolchain, focusing on heterogeneous linking; subsequently optimized Caffe operators for MLU accelerators.		
Megvii	Algorithm Engineer Intern	2019
Optimized traffic video understanding algorithms and productionized them in C++.		
Microsoft Asia Engineering Academy	Software Engineer Intern	2016
Evaluated LevelDB, Presto, and Hadoop for advertising data storage and analytics workloads.		

Selected Publications

Efficient LLM Inference on ARM via Hardware-Aware Operator Co-Design and Heuristic Mixed-Precision Search	ISCAS	2026
RecGPT-Mobile: On-Device Large Language Models for User Intent Understanding in Taobao Feed Recommendation	SIGIR	2026
MNN-LLM: A Generic Inference Engine for Fast Large Language Model Deployment on Mobile Devices	MM Asia	2024
Walle: An End-to-End, General-Purpose, and Large-Scale Production System for Device-Cloud Collaborative Machine Learning	OSDI	2022
LISA: Locality-Informed Speculative Decoding for Accelerating Autoregressive Image Generation	ECCV	2026
ARC-Decode: Accelerated Decoding with Risk-Bounded Acceptance	ICML	2026
Prism-MoE: Efficient Dense-to-MoE Conversion for Visual Autoregressive Generation	ICML	2026
AccKV: Towards Efficient Audio-Video LLMs Inference via Adaptive-Focusing and Cross-Calibration KV Cache Optimization	AAAI	2026
VertiKV: Vertical-Integrity KV Cache Compression for Efficient Multimodal Long-Context Inference	ACM MM	2026
MadaKV: Adaptive Modality-Perception KV Cache Eviction for Efficient Multimodal Long-Context Inference	ACL	2025

Patents & Applications

BANG-Linker: Heterogeneous Linking Method, Apparatus, and Related Products	Cambricon · Filed	2019
BANG Simulator: Programming Debugging Method, Apparatus, and Related Products		
Solution for Deploying Large Language Models on Mobile Devices	Alibaba · Filed	2024
Saliency-Aware Blockwise Quantization via Floating-Point Zero-Point Degrees of Freedom	Alibaba · Pending	2026
Mixed-Precision Allocation for Low-Bit LLMs Based on Action-Conditioned Operator Error		

Honors & Competitions

IEEE AICAS Grand Challenge: LLM Software and Hardware System Co-optimization	First Prize	2024, 2025
NPU Competitions: Tecorigin Operator Development & SOPHGO TPU Programming	First Prize	2022, 2024

Skills

Languages	C, C++, Python, Assembly, CUDA, OpenCL, Vulkan, Metal, BANG, Java, Objective-C
Expertise	Compilers, AI inference engines, CPU/GPU/NPU operator optimization, model conversion, quantization, large language models